

M Sai Shanmukha Jayanth

mjayanth0519@gmail.com • +91 9392696300 • [LinkedIn](#) • [GitHub](#) • [LeetCode](#)

Machine Learning · Python · XGBoost · TensorFlow · CNNs · Statistical Analysis · SQL

SUMMARY

Third-year CS undergrad (GPA 8.94) with hands-on experience building end-to-end ML pipelines, predictive models, and computer vision systems across retail, logistics, and real estate domains. Demonstrated 56% to 97% model accuracy improvement via transfer learning; seeking a data science or machine learning internship to apply statistical analysis, exploratory data analysis, and deep learning skills to real-world business problems.

EDUCATION

Shiv Nadar University, Chennai

Aug 2023 – May 2027

B.Tech in Computer Science and Engineering • GPA: 8.94/10 (through 4th semester)

Narayana Junior College, Hyderabad

July 2021 – May 2023

Class XII • 98.6%

SKILLS

Languages Python • SQL/MySQL • C • Java • C++

ML / DL TensorFlow • Keras • PyTorch • Scikit-Learn • XGBoost • Pandas • NumPy

Specialisations Exploratory Data Analysis • Statistical Analysis • Transfer Learning • Computer Vision • NLP
• Generative AI

Tools Git/GitHub • Jupyter • Google Colab • Matplotlib • VS Code

WORK EXPERIENCE

Machine Learning Intern

July 2025 – Jan 2026

Labmentix

- Delivered end-to-end ML pipelines across 4 business domains (retail, logistics, real estate, customer feedback), standardising SQL-based multi-dataset joins into reusable modules version-controlled via Git.
- Improved delivery-time prediction accuracy by approx. 18% (RMSE on held-out test sets) using XGBoost with 5-fold cross-validated hyperparameter tuning across 50k+ logistics records; surfaced results via Matplotlib visualisations.
- Boosted model performance by up to 12% through statistical feature engineering — geospatial distance metrics, lag/rolling-mean time-series variables, and VADER sentiment scores — validated via EDA and correlation analysis.
- Increased aerial image classification accuracy from 56% to 97% (F1-score) by replacing a baseline CNN with MobileNetV2 transfer learning and optimised tf.data pipelines, cutting training time by approx. 40%.

PROJECTS

Aerial Object Detection

Sept 2025

97% accuracy classifier • MobileNetV2 transfer learning • TensorFlow

- Increased classification accuracy from 56% to 97% using MobileNetV2 transfer learning; validated with confusion matrix, precision, recall, and F1-score across 5 classes.
- Reduced training time by approx. 40% via tf.data caching and prefetching; improved generalisation through targeted augmentation strategies.

Restaurant Clustering Analysis

July – Aug 2025

K-Means segmentation • NLP sentiment scoring • PCA (95% variance retained)

- Identified sentiment and service quality as primary satisfaction drivers via statistical analysis (t-tests, Pearson correlation) and EDA on a unified multi-source dataset.
- Delivered Silhouette Score-validated customer segments via K-Means on PCA-reduced features, enabling data-driven marketing strategy recommendations.

Integrated Retail Sales Forecasting

Aug – Sept 2025

XGBoost regressor • Time-series feature engineering • Inventory optimisation

- Engineered lag and rolling-mean time-series features capturing sales memory and short-term trends, increasing weekly forecast accuracy at store-department level.
- Built an XGBoost regressor on economic and promotional signals to support retail inventory optimisation and staffing allocation decisions.

CERTIFICATIONS — DEEPLARNING.AI / COURSERA

- Supervised Machine Learning: Regression and Classification — May 2025
- Neural Networks and Deep Learning — June 2025
- Improving Deep Neural Networks: Hyperparameter Tuning, Regularisation and Optimisation — July 2025